

ANAND PAGNIS

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EDUCATION

Worcester Polytechnic Institute
Bachelor of Science, Computer Science

May 2026
Worcester, MA

SKILLS

- **Backend & APIs:** Python, REST APIs, FastAPI, Async Workflows, PostgreSQL, Docker, Microservices
- **AI & ML:** TensorFlow, Feature Engineering, Model Evaluation, Backtesting, NLP, Recommender Systems, RAG Pipelines
- **Frontend:** React, Streamlit, Typescript, CSS
- **Languages:** Python, Typescript, C/C++, HTML, SQL

EXPERIENCE

Buffeat

Sep 2025 - Nov 2025

Software Engineer – Frontend Developer

Remote

- Developed and maintained responsive user interfaces for Buffeat's core platform using React and Typescript and standardized branding policies.
- Implemented reusable component libraries and state management solutions, reducing development time for new features by 20%.
- Participated in Agile sprints, including sprint planning, daily stand-ups, and code reviews, delivering features on time with high code quality.

Fundesteam

Aug 2024 - Oct 2024

Project Lead & System Administrator

Panama City, Panama

- Partnered with the Ministry of Education to develop a low-power, scalable, offline educational server for rural Panama.
- Achieved 85% faster video load times through an iterative development process using Docker.
- Automated course uploads with Bash scripts and optimized OS performance, reducing CPU temperatures by 13%.

LEADERSHIP

Investing Association

Jan 2025 - Mar 2026

President

- Led discussions on financial markets, investment strategies, and risk analysis while organizing campus-wide events for a 250+ member club and managing a \$10,000 budget.
- Launched a six figure student led investment fund and served as a voting member on its committee.

Alpha Chi Rho

Aug 2024 - May 2025

Social Chair

- Led a 10-member committee to organize professional and social events for 100+ students, strengthening brotherhood connections on campus.

PROJECTS

ADArC : Automated Design Tool for Arduino Circuits

Aug 2025 - May 2026

Worcester, MA

- Expanded an Arduino circuit simulator by 25+ components using TypeScript, Blazor, AVR8js, and GraphSynth.
- Implemented I2C, SPI, UART, RS485, analog, and digital sensor/actuator simulation with real-time SVG visualization.
- Built Arduino code generation support with dynamic pin mapping and custom library integration.
- Developed event-driven synchronization between UI, simulation controllers, and AVR microcontroller execution.

SEC 10-K Research Terminal

May 2026

- Built a full-stack financial research platform Python/FastAPI backend and React/TypeScript frontend, with type safety across the REST API, eliminating data mismatches across 15+ models from SEC EDGAR to the user interface.
- Implemented hybrid RAG retrieval system combining BM25 keyword matching with nomic-embed-text dense vectors, reciprocal rank fusion, and lightweight reranking, reducing top-3 retrieval error rate by 40%.
- Developed heuristic SEC document parser that extracts sections, financial statements, and XBRL metrics from raw 10-K HTML, applying title normalization and currency/percent cell merging to produce structured, citation-trackable outputs with stable evidence IDs.

Hospital Application

Jan 2024 - Apr 2024

WPI, Brigham and Women's Hospital

Worcester, MA

- Led development of a hospital navigation and service management app as Project Manager, coordinating a 9-member team using Agile methodologies and tools like Jira.
- Deployed website infrastructure with AWS EC2, Docker, Prisma ORM, and backend database using PostgreSQL.